

PHYS 455 Examination

Quantum Mechanics and Wave Functions

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
1. In the context of the uncertainty principle, what is the minimum value of the product $\Delta x \cdot \Delta p$ for a quantum particle?

(A) $\hbar$
(B) $\hbar/2$
(C) h
(D) $h/4\pi$
 2. For a particle in a one-dimensional infinite square well of width L , which statement correctly describes the energy eigenvalues?

(A) They are evenly spaced with separation $\hbar^2/2mL^2$
(B) They scale as n^2 where n is the quantum number
(C) They are independent of the well width L
(D) They form a continuous spectrum
 3. What physical interpretation does the Born rule provide for the wave function $\psi(x, t)$ of a quantum particle?

(A) $|\psi(x, t)|$ represents the probability density
(B) $\psi(x, t)$ represents the particle's definite position
(C) $|\psi(x, t)|^2$ represents the probability density for finding the particle at position x
(D) $\text{Re}(\psi(x, t))$ represents the observable position
 4. Which property distinguishes a Hermitian operator in quantum mechanics and makes it suitable for representing physical observables?

(A) It always commutes with the Hamiltonian
(B) It has real eigenvalues and orthogonal eigenfunctions
(C) It preserves the norm of wave functions under time evolution

(D) It generates unitary transformations exclusively

5. Explain the physical significance of commutators in quantum mechanics. If two observables have a non-zero commutator, what does this imply about simultaneous measurements of these observables?
6. Describe the quantum harmonic oscillator and explain why it serves as a fundamental model in quantum mechanics. What are the key features of its energy spectrum and wave functions?
7. State the time-dependent Schrödinger equation and explain its role as the fundamental equation of motion in quantum mechanics. How does it differ from the time-independent version, and when is each appropriate?
8. Explain the concept of quantum tunneling. How does it arise from the wave nature of particles, and what are its implications for particles encountering potential barriers that would be classically insurmountable?